



From Rhythm to Recognition: “Leveraging Deep Learning and Machine Learning Models for Music Genre Classification”

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Declaration

**‘I declare that this Applied Research Project that I have submitted to Dublin Business School for the
award of [name your programme here] is the result of my own investigations, except where
otherwise stated, where it is clearly acknowledged by references. Furthermore, this work has not
been submitted for any other degree.’**

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Abstract

Music powerfully reflects and shapes human culture, yet the task of classification of music into distinct genres is inherently challenging given the existence of stylistic overlaps, cultural specificity, and the subjectivity of listener interpretation. This research explores the evolving landscape of music genre classification through application of machine learning and deep learning techniques, with a specific focus on the implementation and performance evaluation of Convolutional Neural Networks (CNN), Support Vector Machine (SVM) and Random Forest (RF) algorithms. An extensive evaluation of 30 CNN architectures were carried out, along with 5 SVM and 5 RF model configurations using large datasets like FMA-large and MuMu datasets. Feature extraction techniques – particularly Mel-frequency Cepstral Coefficients (MFCCs) and Mel-spectrogram were determined to be highly significant in distinguishing the acoustic feature specific to each genre.

The research addresses persisting challenges to genre classification, such as class imbalance issues, the emergence of hybrid genres, and questions of cultural representation. Through empirical testing and architectural improvements, the research determines strong model configurations that outperforms conventional approaches on genre recall and macro-average F1 score metrics. This research contributes to the development of genre classification systems that are not only technologically adept but also contextually through, offering important implications for applications in music streaming, recommendation systems, and global music archiving.

Introduction & Overview

Music is one of the most enriching forms of human culture with artistic, emotional, and social aspects that cross linguistic and geographic borders. With the exponential growth of digital music libraries and popularity of streaming websites, being able to efficiently organize, retrieve, and recommend music has become a crucial necessity for music information retrieval (MIR) systems. One of the fundamental problems within this field is that of music genre classification-the automatic identification of audio tracks into stylistically similar genres like jazz, classical, electronic or rock. Music genre classification is a difficult computational problem even though it seems so simple because musical genres are dynamic, overlapping and generally subjective (**Tzanetakis & Cook, 2002; Schindler & Lidy, 2016**).

The traditional approach to genre classification was based predominantly on human annotation and rule-based systems with low scalability and consistency. With the evolution of artificial intelligence, machine learning (ML) techniques came to dominate with statistical learning algorithms employed to model genre boundaries in term of low-level acoustics features like tempo, timbre, pitch and rhythm (**Lidy & Rauber, 2005**). Traditional models like Support Vector Machine (SVM) and Random Forest (RF) provided encouraging performance, especially when coupled with hand-engineered features like Mel-frequency Cepstral Coefficients (MFCCs), chroma vectors, and spectral contrast (**Tzanetakis & Cook,2002; McFee et. al. 2015**). These models would, however fail to capture the complexity and temporal dependencies involved in musical compositions, and therefore did not generalize well on varied datasets.

The advent of deep learning (DL) techniques, especially Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) transformed the genre classification tasks through the ability

of learning hierarchical representation directly from raw or minimally processed audio signals (**Choi et. al. 2017; Panigrahi et. al. 2023**). CNNs, which were initially developed for image classification, have also been successful at audio classification because they can extract spatial and temporal features from spectrograms – two-dimensional plot of sound over time (**McFee, 2015**). Furthermore, architectures involving CNNs over RNNs, like Bidirectional Long Short Memory (Bi-LSTM) have been suggested in order to model sequential properties of music and enhance classification robustness (**Mesghali & Askari, 2022**).

This research explores deep and traditional machine learning models for music genre classification based on two well known and different datasets: FMA large dataset containing metadata and audio files of thousands of tracks and the MuMu dataset with rich genres and comprising multilingual and cross-cultural music tracks. The study conducts a systematic experimental procedure that entails building and assessing multiple diverse CNN architectures, different in depth, filter configuration, dropout rates, and optimization techniques. Meanwhile, SVM and RF models are built and experimented on with both traditional metadata features and extracted audio representations.

A critical part of the research project is the feature extraction pipeline that converts raw audio data into meaningful representations that can be used as model input. This entails using Mel-Spectrograms, Mel-frequency Cepstral Coefficients (MFCCs), and one time-frequency transforms that preserves music perceptual characteristics like tone, rhythm, and harmonic content (**Tzanetakis & Cook, 2002; Lidy & Rauber, 2005**). These are especially important features to CNN-based models as they enable the architecture to learn applicable patterns that relate to various musical genres.

Another major focus of the research is placed on preprocessing and augmentation methods. Methods like trimming silence, audio normalization, and pitch shifting are used for enhancing the quality and balance of the training data. For addressing class imbalance often present with genre classification because of an irregular distribution of music genres, class weighting, stratified sampling, and focused augmentation are used, thus allowing for unbiased model assessment and improving minority genre representation (**Balkrishna & Singh, 2022**).

The research is designed to provide a comprehensive comparative assessment of Convolutional Neural Networks (CNNs), Support Vector Machine (SVM) and Random Forest (RF) models. Every methodology is evaluated on criteria not only on its classification effectiveness, but based on its generalizability, interpretability, and ease of usability. The models are implemented using platforms including TensorFlow, Keras, and Scikit-learn's, while extraction of features is carried out with the help of Librosa package (**McFee, 2015**). The experimentation setup is designed with an emphasis on evenness in datasets, standardized evaluation metrics, and reproducibility among all compared models.

This research project makes a contribution to music information retrieval through a detail analysis of model architecture, preprocessing processes, and respective genre-classification strategies. Through systematic comparisons of deep learning methodology with traditional models across multiple datasets, this research provides insightful assessments of inherent merits and demerits of each methodology. Finally, the work concludes with the identification of optimized Convolutional Neural Network (CNN) architectures supported by comparative data, which holds significant potential for use within music recommendation programs, audio search engines, and digital library organizations.

Literature Review

Music Genre classification is an essential task in the field of music information retrieval designed to automatically classifies songs based on musical compositions by genres such as rock, hip-hop, pop or jazz. The genre can be viewed as a semantic label denoting common musical features, contextual associations and listeners expectations (Elbir et.al.,2018). In the context of MIR, these labels enable effective retrieval systems, personalized recommendations and organized administration of vast digital libraries of music (Panigrahi et. Al.,2023). Traditionally, genre classification heavily relied on manual annotation by musicologists or metadata tagged by all record labels, subject to inconsistency and personal judgements.

Emergence of streaming audio platforms such as Spotify and Apple music has led to extensive growth in the amount of availability of music content. Uploading and streaming of millions of tracks on daily bases has made manual classification virtually unattainable. This requirement of scalable processes has accelerated the shift towards the automation through Machine Learning (ML) and Deep Learning (DL) (Yuwono et. Al.,2023). Auto genre classification plays a crucial role in operations of advance recommender systems, contextual playlist creation and improving streaming platform engagement. (Soni & Kalra,2023).

Early ML approaches were founded on algorithms like Support Vector Machines (SVM), k-Nearest Neighbours (KNN), and decision trees that employed hand-engineered features like tempo, Mel-frequency cepstral coefficients (MFCC), and spectral roll-off to represent musical properties. (Yuwono et. Al.,2023; Elbir et.al.,2018). Even though these models were quite effective, they often struggled with ambiguity where there were hybrid genres and genre crossovers. Their reliance on featured sets predefined by humans limited their ability to evolve to upcoming music styles and cultural diversity. (Jiang et al., 2024).

With the advancement in deep learning, and particularly in areas of Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) has greatly facilitate genre classification abilities. Advance deep models have the ability to automatically extract hierarchical features of both spectrograms and raw audio, to identify subtle complex patterns that are tend to be disregarded by traditional algorithms. (Pelchat & Gelowitz, 2024; Panigrahi et al., 2023). CNN, in particular have proved high accuracy in classifying genres with analyzing time-frequency patterns. (Sharma et al. 2021). Also using Bi-LSTM models to extract temporal dependencies has further enhanced genre recognition. (Mesghali & Askari, 2022).

Today, musical genre classification moves towards hybrid models, collaborative approaches, and collections reflective of cultural diversity. This development attempts to address prior limitations and aligns with the border goal of intelligent and personalized musical experiences (Gao & Liu, 2023). The concept of 'Music Genre' is inherently malleable and complex. Broadly, a music genre refers to an identifier for categories of musical pieces differentiating by shared stylistic traits, including instrumentation, rhythm, harmony, lyrics and the cultural contexts of origin. (Elbir et.al.,2018). Genre not only serves as descriptive terms but are useful tools in the areas of music retrieval and recommendation systems. Nonetheless, the definition of an unambiguous description of the genre is too challenging to achieve since the growing blurring of genre margins, artistic creativity, and regional influence tend to undermine classic categorizations. (Jiang et al., 2024). The ambiguity in genre classification, introduces inconsistency between different datasets and provide major challenges to automated classification systems. (Panigrahi et al., 2023).

Earlier computational music genre classification methodologies relied largely on the extraction of handcrafted acoustic features from audio signals. These include the use of Mel-Frequency Cepstral Coefficients (MFCC), which mimic how the human auditory perceives pitch and timbre; spectral roll-off, which is a measure of the energy below given percentage of the total energy of the signal; and zero-crossing rate, which is the rate at which the signal crosses zero. (Elbir et.al.,2018; Sharma et al. 2021). Other features, including tempo, chroma and spectral centroid, were also used to capture the rhythmic and tonal characteristics that are native to music.

Following feature extraction, the traditional machine learning (ML) classifiers were employed to assign genre labels. Some of the commonly used models include Support Vector Machines (SVM), k-Nearest Neighbours (k-NN), Naïve Bayes, and Random forests. SVMs were particularly prevalent due to their ability of handling high-dimensional data, as well as for their performance in cases where linear separability is needed. (Yuwono et. Al.,2023). On the other hand, k-NN classifiers identified class membership through proximity in the feature space. These models were traditionally trained and tested on datasets including the GTZAN, ISMIR and the million-song dataset.

While these traditional approaches were a significant improvement over manual tagging but suffered from several critical limitations. Firstly, the performance of manually designed features was highly reliant on factors like recording quality, acoustic environment, often failing generalization when applied to varied datasets. (Pelchat & Gelowitz, 2024). Secondly, the reliance of a fixed features set limited the capacity of the model's ability to adapt to the evolving characteristics of music, especially when it came to classify non-western and experimental forms of music. (Gao & Liu, 2023). Lastly, traditional classifiers, such as SVMs struggled with overlapping genre boundaries, where a single composition of music can represent traits from several different music genres. (Panigrahi et al., 2023).

The key limitation was lack of hierarchical feature learning. Traditional models were not able to learn abstract feature independently from raw audio data, and therefore needed to depend on domain knowledge during the feature engineering phase. This often led to models becoming susceptible to overfitting the specific datasets but failing to generalize well under real world settings. (Sharma et al. 2021). Moreover, genre labels at times contain noise and inconsistencies, further worsening the problem faced by rigid classification algorithms. These challenges catalysed the driving forces behind the switch to deep learning approaches that supported end-to-end learning from raw or unprocessed data and enhance the ability to handle sophisticated audio patterns. The shortfall traditional machine learning laid the basis on which to further develop genre classification during the deep learning period.

The domain of music genre classification has recently witnessed an overhaul with the uses of deep learning methodology. As opposed to traditional machine learning techniques reliant on hand engineered features, deep learning architecture are able to learn hierarchical representations from raw and pre-processed datasets, improving the accuracy of classification and generalization. Among the various deep learning architectures, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), mixed models combining CNNs and RNNs, and recent improvements in transformers and autoencoders, have shown high performance.

Convolutional Neural Networks (CNNs) are the focus of much attention among the realms of image recognition tasks on account of their ability to determine spatial hierarchies using convolutional layers. Regarding music genre classification, audio signals are often converted to spectrograms, which are the representation of the audio data in the time-frequency domain. Spectrograms are equivalent to grayscale image and, as such, are very effective in CNNs to determine localized as well as global

spectral features. Convolutional Neural Networks (CNNs) have been shown to excel at recognizing the unique frequency and temporal signatures of different musical genres. The Convolutional filters are capable of identifying pitch contours, rhythmic textures, harmonic patterns, and dynamic variations, which are essential characteristics of genre. **(Panigrahi et al., 2023)** focused on applying CNNs to GTZAN dataset in their work, achieving scores of close to 79% through spectrogram analysis.

Different CNNs architectures are explored specifically for the given endeavour. The VGG architecture-based networks, initially created for the ImageNet corpus, are adapted for spectrogram-based inputs. These networks use multiple stacked convolutional layers along with the use of the pooling operation to reduce spatial dimensions without losing the significant features. **(Pelchat & Gelowitz, 2024)**. ResNet architectures, where the use of residual connections in the hallmarks, have also been adapted to overcome the vanishing gradient problem in deep networks.

In some cases, applying 1D CNNs directly on audio feature vectors such as Mel-Frequency Cepstral Coefficients (MFCC) as opposed to 2D spectrograms. This approach helps in reduction computational complexity and enables the model to capture temporal correlations in lower-dimensional audio descriptors. Convolutional Neural Networks (CNNs) are proficient at recognizing spatial patterns but are restricted in modelling long-term dependencies in sequential data. Music is inherently sequential, containing evolving melodic and rhythmic patterns through time. Recurrent Neural Networks (RNNs) and the LSTM and GRU architectures in particular, are best at addressing time-series data and remembering previous inputs.

RNNs functions through by feeding information from one step to the next in time, enabling the model to acquire transition knowledge. The vanilla RNNs are improved on by the LSTMs through mitigating vanishing gradients problem through gated recurrent units that manage the flow of information. The GRUs, which are less complicated than the LSTMs, do the same but with fewer parameters. **(Mesghali & Askari 2022)** demonstrated the capability of Bi-LSTM networks, which consider sequences in forward and backward directions, on the Million song dataset. Their results demonstrated the capability of RNNs and specifically Bi-LSTM architectures in modelling long-term dependencies that exhibit subtle genre change. Another study by **(Wu et. Al. 2022)** as a comparative analysis of GRUs and LSTMs found that the accuracy was slightly higher with LSTMs but GRUs converged faster and were computationally efficient. Their contribution was to decide the applicability of the architecture depending on the complexity of the problem and the computational resources available.

While CNNs excel at spatial modelling and RNNs excel at temporal modelling, many researchers widely suggested to have hybrid architectures that integrate both. The common framework has the spectrogram pass through CNN layers to obtain local time-frequency features and then through RNN layers to capture the dependencies between frames. This CNN-RNN architecture employs CNN for feature extraction and RNN for modelling the temporal. For example **(Soni & Kalra, 2023)** integrated 2D CNN and Bi-LSTM units to classify genre for multilingual music data. Their approach surpassed the performance of separate CNN or RNN models, particularly in identifying genre transition between fusion tracks. In same vein, **(Sharma et. al. 2021)** found that adding the use of temporal modelling in CNN pipelines reduced genre confusion between competing categories such as electronic vs ambient. They also attempt attention layers on top Bi-LSTM output to permit the model to selectively attend to the most informative areas of the sequence. Moreover, end-to-end hybrid models were proposed in a number of configurations: CNN on the spatial encoding followed by LSTM,

GRU or even transformer layers to capture sequence dynamics. These architectures are the state-of-the-art for genre-classifying tasks as of today, with high accuracy using real-world noisy data.

Transformers originally formulated for natural language processing, are increasingly being employed for audio classification since the transformers can model long-range dependencies with the assistance of self-attention. The transformers do not necessarily need to process the input sequentially and can ingest all input positions simultaneously, which significantly enhances training efficiency. As per work of **(Yoon et. al. 2023)** employed a transformer-based genre model that outperformed the standard RNN and CNN-based models on multilingual datasets. The system handled global structural patterns of music well, something that is essential for large-scale theme categories such as Jazz and Classical. Attention mechanisms are now being used to enhance both CNNs and RNNs by weighting different parts of the input sequence or the feature map. Attention allows the model to learn to attend to genre-specific frames or frequency bands. **(Yoon et. al. 2023)** added self-attention to CNN & Bi-LTSM model and achieved improved accuracy over the non-attention baseline. The model, with the attention mechanism, was able to focus on the rhythm breaks in Hip-hop and the harmonics modulation in jazz-moments that are lost in average pooling. Other attention-based designs including transformer-CNN hybrids and multi-head attention modules have exhibited good performance on benchmarking tests.

Autoencoders have also been used for unsupervised learning, have also been used for genre classification. Autoencoders reduces noise and also data dimensionality by learning compressed forms of input sound in form of latent codes. They make subsequent classification more efficient. In a study by **(Wu et al. 2022)**, Variational autoencoders (VAEs) were employed to pre-train mel-spectrogram embeddings that were classified using simple dense network or CNNs. Their results obtained better accuracy rates in comparison to those of from-scratch trained models, particularly in low-resource settings. Autoencoders have also been used in conjunction with clustering algorithms to discover latent genre patterns within unlabelled data to develop a scalable method for semi-supervised classification.

In general, the deep learning facilitates considerable music genre classification advancements. CNN remains good tool for spectral feature captures, while RNNs perform well in temporal dynamics. Blending the two using hybrid techniques continues to set the benchmark for robustness and accuracy. Emerging architecture in form of transformers and autoencoders are also developing in this area, allowing researchers to model global music structure and extract unsupervised features effectively. Attention mechanism based-models and pretraining techniques are also improving accuracy levels for challenging, hybrid and lesser-represented genres.

The effectiveness of music genre classification performance, especially of deep learning-based models, hinges greatly on the representation of musical data. Audio signals are multivariate and complex by nature and contain temporal, spectral and timbral information. It depends largely on a good representation and careful application of feature engineering to have a good classification performance in handling diverse genres and noisy real-world data.

Spectrogram & Time-Frequency Representation.

One of the most common forms of representation used in deep learning for music classification is spectrogram as a two-dimensional graphical representation of the spectrum of frequencies of the signal in the axis time. A popular variant is Mel-Spectrogram, that make use of the Mel scale to have a closer match to human auditory perception. The Mel-scale compresses the higher frequencies more than the lower ones, mimicking how the human ear perceives pitch. Experimental results have proved that directly trained CNNs on Mel-spectrograms offer much higher classification

accuracy in comparison to raw waveform input. **(Choi et.al., 2017)**. Mel-frequency cepstral coefficients (MFCCs), although traditionally been designed in hand-engineered way, are usually extracted from the Mel-Spectrogram and have been widely adopted in both traditional as well as in deep learning pipelines. **(Elbir et. al., 2018)**.

More advanced representations like constant Q-transform (CQT) and wavelet transforms have also been used to provide higher resolutions in the low frequency range. CQT allows for logarithmically scalable frequency, which aligns better in capturing musical pitch perception and harmonica structure. **(Liu et. al.2021)**. Similarly, wavelet transforms also provide multi-resolutions analysis capabilities, enabling the model to learn both the short-duration transients and the long-duration harmonica structures in music. Hybrid model uses both Mel-spectrograms and wavelets as complementary input so to utilize both of these to their advantage. **(Wu et. al. 2022)**.

Learned Representations Vs Hand-Engineered Features

In the traditional approach, music genre classification has relied on hand-engineered features such as Mel-frequency cepstral coefficients (MFCCs), zero-crossing rate, spectral centroid and chroma features. These features are derived from audio signals using signal processing techniques and fed to machine models such as Support Vector Machine (SVM), k-Nearest Neighbours (k-NN) and Random Forest. **(Yuwono et. al.,2023)**. However, with the advent of deep learning, there has been clear shift towards learned representations whereby hierarchical features are automatically learned by neural networks from spectrograms and even raw waveforms. **(Choi et.al. 2017)** demonstrated end-to-end learned Convolutional Neural Networks (CNNs) for spectrogram inputs could outperform traditional classifiers on Mel-frequency cepstral coefficients (MFCCs) without the need for explicit feature engineering. Learned features are more flexible and often more expressive in nature because they can capture genre-dependent features that cannot easily be quantifiable through manual methods. Despite the strength of learned representations, their fusion with domain-specific designed feature has found to improve interpretability and robustness. **(Silla et. al. 2020)** for instance integrated spectral contrast and tempo to alongside Convolutional Neural Networks (CNNs) features to produce better genre prediction performance in hybrid genres. It implies that hand-crafted features still remain beneficial, particularly in biased or low-resource datasets.

Importance of Feature Engineering in Deep Learning

Although deep learning models should ideally automate the process of feature extraction; feature engineering still plays a critical role in the pre-processing and normalization process. The spectrograms must be computed using the same hop lengths and window sizes and must also be normalized across samples to provide stable training. Inconsistent parameters for feature extraction may result in poor generalization or convergence issues. **(Panigrahi et. al. 2023)**. Moreover, the format and scale of features (e.g. dB-scale vs. linear) also influence the degree to which Convolutional Neural Networks (CNNs) can learn. Pre-processing operations like log compression of the spectrograms have already been shown to assist models in picking up on important difference across samples, especially for lower-energy frequencies **(Kim et. al. 2021)**.

Data Augmentation Techniques

To enhance generalization and reduce the risk of overfitting, especially when working with datasets that has class imbalance, researchers utilize data augmentation techniques approaches. Augmentation music data involves.

- Pitch Shifting – Alters the pitch but not tempo
- Time-Stretching – Modified the rate without modifying the pitch
- Noise Injection – Adds whites or pinks noise to stimulate real conditions
- Dynamic Range Compression -Alter loudness contrast to enhance model robustness.

(Choi et. al., 2019) used multiple augmentations to utilize while training and achieve 4-7% absolute accuracy gain on a number of datasets. Similarly, (Muller & Ewert 2020) suggested that model-based augmentation methods can make a model invariant to small distortions and hence more robust against aural playback conditions and recording properties. Another method involves the use of multiple representations in the training. For instance, combination raw waveform Convolutional Neural Networks (CNNs) and Mel-Spectrogram-based Convolutional Neural Networks (CNNs) together in ensemble model has improved performance because the model gets the best of both the spectral and the temporal features. (Panwar et. al. 2023).

Despite significant advances in music deep learning and music information retrieval (MIR), music genre classification remains afflicted by several major problems. These challenges stem not only due to technical difficulty of representing the sound signals but also due to the natural subjectivity and cultural richness of the music. Key issues include handling hybrid genres and genre overlap, addressing cultural diversity and dataset bias, handling real-world noise in the sound signals, and considering the listener's subjective experience.

Handling Hybrid Genre and Genre Overlap

One of the primary difficulties in music genre classification is the intersectionality of genres and the formation of hybrid genre. Unlike traditional classification tasks that typically entail mutually exclusive categories, music genre tends to have overlapped stylistic, instrumental, or rhythmic characteristics. For instance, genres such as Pop, Rock, Jazz, Fusion, or Electronic Dance Music (EDM) blend aspects of various genres, which makes it difficult for classifiers to identify distant categories effectively. (Tzanetakis & Cook 2002), assert that numerous songs display characteristics from multiple genres concurrently, thereby contesting the notion of genre purity that serves as the foundation of various classification systems.

(Panigrahi et. al.2023) argued that Convolutional Neural Networks (CNNs) models trained on spectrogram features can struggle to distinguish between the Rock and Metal genres due to overlapping frequency patterns and instrumentation. Similarly, (Yuwono et. al. 2023) found that Support Vector Machine classifiers faced difficulties when genre boundaries were unclear, with a tendency to misclassify tracks into neighbouring categories. Such overlaps reduced the accuracy of multi-class models and demonstrate the necessity for multi-label classification frameworks, in which songs are allowed to belong to multiple genres at once.

Addressing Cultural Diversity and Bias in Datasets.

Another critical concern involves lack of sufficient cultural diversity in datasets, which therefore creates bias and limits generalizability. Many of the benchmark datasets like GTZAN and FMA are predominantly western in scope, focusing on Rock, Pop, Jazz and Classical music styles. This poor representation of non-western genres like Qawwali, Afrobeats or Carnatic demeans the ability of model to generalize well across different genres of global music (Gao & Liu, 2023). (Jiang et. al. 2024) ran an analysis using Spotify streaming data and revealed the way that cultural tastes affect genre boundaries differently across regions. They noted that failure to include wide arrays of cultural contributions can

lead systems of classification to mislabel or overlook entirely specific music traditions. This can perpetuate algorithms bias and decrease the performance of music recommendation systems within multicultural settings. Efforts to mitigate such bias involves the choice of well-balanced multi-ethnic datasets and the utilization of language-independent features like rhythm and pitch contour. Transfer learning has also been suggested as a strategy for adapting models to culturally diverse but sparsely populated datasets. **(Wang & Song, 2023)**.

Dealing with Noisy and Real-world Data

While most classification systems are train to studio-quality audio, real-world audio and clean is usually noisy or heterogenous. Low fidelity recordings, background noise, crowd noise in live performances, and mastering differences produce acoustic distortions that drastically degrade model performance. **(Pelchat and Gelowitz 2024)** noted that their CNN models, while robust under idealized conditions, broke down against real-world noise, especially in user generated music and live tracks. To combat this, various data augmentation techniques like noise injection, pitch shifting and time stretching are used to stimulate real-world environments during training of the models. **(Kim et.al,2021)**. Nonetheless, it continues to be a challenge to fine-tune models to be robust in uncertain audio environments. Moreover, same genres such as Punk or Lo-Fi, Hip-Hop intentionally embrace distortion and low fidelity as stylistic choices, further complicating model's assumption about 'noise' vs 'style'.

Capturing Subjective & Contextual Information

The most elusive challenge of music genre classification is probably dealing with the subjectivity of music perception and its variation across contexts. Genre classification is not always grounded in musical form, but is influenced by cultural narrative, emotional tones, and listener's identity **(Soni & Kalra, 2023)**. For instance, same song might be classified as "Electronic" in one society and as "Experimental" in another, relative to social and historical context. Emotional content adds another layer of complicating element. Some genres like Soul or Ambient are likely to be associated with specific moods, but emotion in personal and varies individually. Model trained on strict genre classification may thus fail to capture these differences and make oversimplified predictions. Some researchers have tried to incorporate listener preference data, such as user tags or listening history, into genre classification system. **(Dhanush et. Al. 2023)**. Others work with multi-modal systems that encompass audio along with metadata, lyrics or cultural context in a bid to more fully know genre identity. Nonetheless, to completely explain the semantic and emotional content of music is an open question, one that straddles technological and philosophical borders.

The success of music genre classification models greatly relies on the quality and variability of the datasets that are utilized for training and testing. A number of benchmark datasets have become widely utilized in community, with varying characteristics and difficulties. Examples like GTZAN, Free Music Archive (FMA) dataset or AcousticBrainz dataset. One of the oldest and most popular datasets is GTZAN genre collection, which was first proposed by **(Tzanetakis & Cook 2002)**. It consists of 1,000 audio segments, each 30 second long, and each uniformly distributed within 10 genres; Blues, Classical, Country, Disco, Hip-Hop, Jazz, Metal, Pop, Reggae and Rock. While that dataset has served the baseline reference, it has significant shortcomings. For instance, certain songs are wrongly named, duplicated, or from the same source, which can lead to inflated accuracy scores **(Strum, 2013)**. Additionally, the dataset lack of cultural diversity and with a strong bias towards western music genres.

The Free Music Archive (FMA) dataset, collected by **(Defferrard et. Al. 2017)**, addresses some of the limitations of GTZAN by providing open-source, legally shareable music files from Free Music Archive website. It consists of four subsets small (8000 tracks), medium (25000 tracks), large (106000+ tracks), and full annotated with a total up to 161 genres. FMA allows multi-label classification and includes rich metadata like artist, album, and track-level information. Its large coverage and genre diversity, qualify it as a candidate to train deep learning models. However, FMA is still skewed towards western music and might fail to cover global genre variance.

AcousticBrainz, a collaborative effort between universities and the MusicBrainz community, offers a vast corpus of low-level and high-level music descriptors from the Essentia audio analysis library. In contrast to GTZAN or FMA, AcousticBrainz contains millions of tracks with features such as BPM, key, danceability, and genre tags. It's good choice for application other than classification, such as emotion recognition and music similarity. However, genre labels in AcousticBrainz are usually crowdsourced and thus making them susceptible to inconsistency and subjective bias. **(Bogdanov et.al.2016)**. In comparison these datasets, FMA is unique in its size and machine learning applicability, GTZAN for its historical role and simplicity, and AcousticBrainz for feature density. None of the three, however, are well-represented on a global scale, which highlights the need for culturally inclusive datasets for future research. **(Gao & Liu, 2023)**.

Evaluation Metrics in Music Genre Classification.

To assess the performance of music genre classification models, several metrics are commonly used:

- Accuracy measures the percentage of tracks that are classified correctly. Although useful as a general metric, in unbalanced datasets it becomes meaningless because large genre can dominate the results.
- Precision measures the ratio of true positives to all positive predictions. High precision suggests a model's trustworthiness in predicting a genre without a false alarm.
- Recall measures how well the model learns to cover all instances of the genre. Recall is essential in guaranteeing that the model achieves completeness, particularly for low-sampled genres.
- F1-score is the harmonic mean of precision and recall. It is most useful for evaluating performance on unbalanced datasets, where false positive and false negative trade-offs are most critical.
- AUC-ROC (Area under the Receiver Operating Characteristic Curve) gives an idea of the model's performance across different classification thresholds, especially for multi-label or probabilistic predictions. Together, these measures provide a balanced perspective of classification performance, enabling researchers to identify trade-offs and tune models to fit practical applications.

Current music genre classification trends are being shaped by several technological, cultural, and methodological trends. One of the significant directions is the creation of robust classification systems that function well in the presence of noisy, real-world settings. There is growing tendency to test models on datasets comprising live performances, user-generated data and non-studio quality recordings that require algorithms to be resilient to background noise and recording quality variance. **(Pelchat & Gelowitz, 2024)**. Data augmentation, noise filtering, and robust loss functions are being employed to enhance model generalization. Another important emphasis is the cultural diversity of

genre classification. Traditional datasets such as GTZAN and FMA are characterized by a strong western genre bias, and music genres in Asia, Africa and Latin America are tend to be neglected. **(Gao & Liu, 2023)**. To alleviate this bias, researchers are advocating the development and utilization of culturally diverse annotated datasets that are capable of holding global music genres. The inclusion of region-dependent genres providing greater representation and enable music genre classification systems to reflect the richness of world music cultural diversity.

The field is moving in the direction of taking advantage of larger and more varied datasets, e.g. AcousticBrainz and Million Song datasets, that have both genre labels and rich metadata. Such datasets enable transfer learnings and multi-task learning, where genre classification is one task among many e.g., mood detection and instrument recognition. **(Wang & Song, 2023)**. In terms of personalization, researchers are incorporating contextual and user-centered feature into classification models. This involves leveraging lyrics, artist data, album review, and listening behavior of users to enhance accuracy and enhanced user experience for recommendation systems. **(Mesghali & Askari 2022; Soni & Karla, 2023)**. These methods go beyond audio content to more effectively identify user intent and song context.

Finally, emerging deep learning architectures such as transformers and attention-based models are gaining traction. Such models, which have been effective for natural language processing, are being formulated for music genre classification by enabling systems to attend to relevant section of music sequences, thereby improving classification performance for genre with complex temporal dynamics **(Yoon et. Al. 2023)**.

Methodology

This research adopts an experimental and comparative approach to compare the performance of multiple machine learning and deep learning algorithms in identifying music genres. The design is centred around testing and training multiple models for Convolutional Neural Networks (CNNs), Support Vector Machine (SVM) and Random Forests (RFs) on two large datasets; the Free Music Archive (FMA) large dataset which includes FMA large audio files and FMA metadata and MuMu dataset. These datasets are selected due to their large sizes, variety of genres, and real-world coverage of music audio and therefore suitable to utilize model performance benchmarking for genre classification. The approach is suitable to this research context as it allows comparative and controlled model architectural evaluation against pre-defined evaluation metrics like accuracy, precision, recall and F-1 score. Deep architectures, and consequently Convolutional Neural Networks (CNN) architectures, are leveraged to identify sophisticated time-frequency characteristics in audio spectrogram representations. **(Panigrahi et. al. 2023)**. Support Vector Machine (SVM) and Random Forests (RFs) architectures are utilized for comparative purposes to allow tangible insight into how deep architectures outperform, or complement, traditional approaches under various experimental conditions.

In the current study, the independent variable includes different classification methods. (e.g. CNN, SVM and RF). Different audio feature representation such as Mel-Spectrogram or Mel-frequency cepstral coefficients (MFCC) and the datasets used are mainly FMA large and MuMu datasets. The dependent variables is classification accuracy, along with relevant measures like precision, recall and F-1 scores. A rigorous methodology has been followed to address the research question with accuracy:

1. Can the identification of subgenre improve music recommender systems? 2. Which deep learning model has better effectiveness in music genre classification? 3. How does feature representation and dataset diversity impact the performance of the models? Each model is trained on the same subset of pre-processed audio data, tested under homogeneous cross-validation methods, and measured using standard metrics. This strict methodological process enables to evaluate model performance under the same experimental conditions. Control measures include consistent data pre-processing for all models, stratified training-validation-testing splits, fixed randomly seeds for reproducibility and hyperparameter optimization using cross-validation. Selective data augmentation methods like noise injection, and pitch shifting were employed to encourage model generalizability. To ensure validity, we performed experiments with de-duplicated, cleaned data with balanced genre coverage where possible. We achieved reliability by re-executing selected experiments to produce repeatable results and by saving model checkpoints and logs to facilitate auditability and transparency. This study leverage two large and publicly available datasets: the Free Music Archive (FMA) large dataset and its corresponding metadata files (tracks.csv & feature.csv) and the MuMu dataset, which consists of Million song metadata and amazon product data. These datasets were chosen due to richness they entail, genre variety, and applicability to large-scale music genre classification tasks.

The FMA large dataset contains 106,574 audio tracks across 161 genres with over 343 GB audio material. All audio files are 30 seconds long and encoded in 16-bit PCM WAV files at the 44.1 kHz sampling frequency. The dataset is annotated under well-formed hierarchical genre taxonomies, although for the needs of the current study, the top 25 genre subset was extracted to focus classification within the most common and well-represented genres. Track information, genre labels, and audio features such as Mel-frequency cepstral coefficients (MFCC), spectral contrast, chroma and tempo were obtained from the track.csv and feature.csv metadata files available.

The MuMu dataset brings additional genre diversity and semantic richness by combining audio samples of the Million song dataset and Amazon product reviews. As such, during the process, the MuMu dataset offered an additional test bed to cross-domain validate model's generalizability. It contains thousands of tracks with 30 top-level genres like Electronic, Jazz, Hip-Hop, International, and experimental marked in MP3 format (**Oramas et. al. 2017**). Subjective classification and commercial categorization are both captured by genre labels, and thus MuMu is very useful for real-world usability study.

Pre-processing of data entails some significant steps. All audio clips were made to have the same level of amplitude to reduce bias caused due to variation in record loudness. Silence removal was conducted using voice activity detection algorithms for removing silent duration portion from the clips. Pre-processing of both datasets was further performed using the Librosa library to achieve Mel-spectrogram representations and Mel-frequency Cepstral Coefficients (MFCCs) as input for the Convolutional Neural Networks (CNN) architectures. The dataset choice is backed by substantial justification. The FMA large dataset is widely used across the research area for genre classification due to its open-access provision, pre-structured class labels, and higher audio quality. (**Defferrard et. al. 2017**). The MuMu dataset is an ancillary dataset supplement prepared to add depth and diversity with commercial context and relevant metadata, for example, user reviews and popularity levels. (**Oramas et. al. 2017**).

Despite their robustness, these datasets have recognizable limitations. The FMA dataset is Western-biased, with insufficient representation of global or traditional generic genres, potentially affecting the generalizability of the trained models. (**Sturm, 2014**). MuMu dataset is also susceptible

to challenges because of its subjective tagging, where genre is based on consumer perception and not to formal musicological conventions. In addition, the 30-seconds length of the FMA clips could leave out important parts of a song that are crucial for the correct identification of genres.

To address these issues, careful data cleaning, augmentation, and rebalancing processes were enforced. Techniques such as pitch shifting, time stretching and noise injection were applied to improve generalization of the models and to prevent overfitting, particularly for less-represented genres. Together, the FMA and MuMu datasets provided complementary perspectives that enabled thorough and realistic inspection of genre classification across diverse music domains.

Feature extraction is a key building block used in the design of efficient music genre classification systems. Feature extraction involves converting raw audio signals into quantitative features that are suitable for analysis with machine learning or deep learning methods. In this study, we focused on extracting audio features that have proven effective in earlier music information retrieval research, Mel-Frequency Cepstral Coefficients (MFCC), Mel-spectrograms, chroma feature, and spectral contrast. The main library that provides an extensive toolset for audio and music analysis.

Audio Feature Extract

The core feature extraction includes:

- Mel-Frequency Cepstral Coefficients (MFCC) – Capture the timbral characteristics of audio and mimic the human auditory perception by emphasizing perceptually relevant frequency.
- Mel-Spectrograms – Mel-spectrograms represents the audio signal power spectral density on a Mel scale, making them suitable for use with Convolutional Neural Network. (CNN)
- Chroma STFT – Chroma STFT characterizes the harmonics and the frequency of the sound using the twelve Western musical scale-based pitch classes.
- Spectral Contrast – Spectral contrast measures the difference between peaks and troughs of the frequency spectrum, thus highlighting the dynamics and texture of musical compositions.
- Zero-Crossing Rate – Zero-crossing rate indicates the rate at which audio signal changes sign, which is useful for distinguishing percussive sounds.

Their choice was determined by their ability to convey a large diversity of musical properties-namely timbre, rhythm, pitch and texture – that are important for genre classification tasks. (Tzanetakis & Cook 2002; Humphrey et. al. 2013).

Feature Extraction Process

Using the Librosa library, audio was digitized with a constant sampling rate of 22,050 Hz. The tracks were made mono and truncated or padded to a constant length of 30 seconds.

- For Mel-frequency Cepstral Coefficients (MFCCs), the following parameters were used:
N_fft = 2048 (window size)
hop_length = 512 (stride)
N_mfcc = 13 (number of coefficients)
Every Mel-frequency Cepstral Coefficient (MFCC) was computed from short-time Fourier transform (STFT), then filtered by a Mel filter bank, and logarithmically scaled.
- Using Mel-spectrograms:

N_mel = 128 Mel bands

Amplitude was scaled logarithmically to facilitate better identification of low-volume frequency. The result was a 2D matrix with dimensions (128, ~1290) for a 30 seconds tracks.

- For Chroma STFT:
n_chroma = 12 pitch classes (C to B)
calculating from the harmonic component of STFT.
- For Spectral Contrast:
7 frequency bands were used, computed from the spectral peaks and valleys.

Each feature matrix was normalized using z-score standardization, thus ensuring a mean of zero and variance of one across the datasets. (Kim et. al. 2021).

To make feature compatible to particular implementations:

3D input dimensions such as (1290, 13, 1) for Mel-frequency Cepstral Coefficients (MFCCs) and (1290, 128, 1) for Mel-Spectrograms were utilized by CNN models.

Both Support Vector Machine (SVM) and Random Forest (RF) models used flattened statistical aggregations – mean, standard deviation, skewness – of the 2D matrices to reduce them to 1D feature vectors (e.g. MFCCs: 2 dimensions x 13 frames = 26 features).

Feature Extraction Failures and Mitigation.

In the processing of over 100,000 FMA large and MuMu dataset audio files, there were some issues that were encountered:

- Inaccessible or damaged files
- Inconsistencies with metadata
- missing or silent audio files

In order to try-expect blocks were implemented to skip and log failures. Arrays of features were aligned with tracks.csv on strict integer-based indexes. Data were filtered to only retain those with complete features and labels. As a result, feature alignment and cleaning steps reduced the usable data set to a ~108,000 valid tracks from original ~136,000 dataset.

```
error
❌ Failed to process /content/drive/Othercomputers/My
Laptop/Downloads/fma_large/155/155320.mp3: [Errno 2] No such file
or directory: '/content/drive/Othercomputers/My
Laptop/Downloads/fma_large/155/155320.mp3'

-----
RuntimeError                                Traceback (most recent call last)
<ipython-input-14-58d0c8c89137> in <cell line: 0>()
      85     continue
      86 if len(X) == 0:
--> 87     raise RuntimeError("❌ No audio samples were loaded.
Please check the dataset path or audio file validity.")
      88
      89

RuntimeError: ❌ No audio samples were loaded. Please check the
dataset path or audio file validity.
```

Mel frequency Cepstral Coefficients (MFCCs) and Mel-spectrograms prove to exhibit high effectiveness since they simulate the human auditory processing and have been thoroughly tested in genre classification and speech recognition tasks. (Logan, 2000; Choi et. al. 2017). Spectral contrast and chroma features are able to capture more complex music features, such as instrumentation and harmony, which are critical for differentiation among jazz, classical, and electronica music genres.

(Tzanetakis & Cook 2002). (Choi et. al. 2017) presented evidence that CNN models learned from spectrogram representations significantly outperform conventional models that rely on hand-designed features. In separate study, (Panwar et. al. 2023) demonstrated that the incorporation of Mel-frequency Cepstral Coefficients (MFCCs) with chroma and contrast features enhances robustness in case of genre overlap.

Libraries & Software's

- Librosa (McFee et. al. 2015); primary tool for audio feature extraction
- NumPy/Pandas: For numerical computation and data frame alignment
- Scikit-learn: For normalization, label encoding and data transformations
- Matplotlib/Seaborn: For visualizing spectrograms and feature distributions.

In conclusion, the careful choice and refinement of auditory features enable robustness in the musically relevant representations acquired by deep learning models enhancing the overall classification performance. The dual strategy of leveraging both hand-crafted and learning features plays an important role in achieving robustness over different genres and datasets.

Deep Learning & Machine Learning Models

1. Convolutional Neural Networks (CNN) Models

Convolutional Neural Networks (CNNs) have emerged as a cornerstone for audio classification tasks due to their ability to automatically derive hierarchical features from time-frequency audio features such as spectrograms and Mel-frequency cepstral coefficients (MFCCs). In this research, a rigorous experimental framework to measure the performance of CNN models in the field of music genre classification tasks using the FMA-large dataset alongside the MuMu dataset, FMA metadata and audio features. The exploration phase of Convolutional Neural Networks (CNNs) began with construction and evaluation of 28 different architectures, each with different configurations, all with the goal of maximizing the effectiveness of music genre classification. The models were developed systematically, with careful tweaks in hyperparameters, structural intricacies, input features, and training models. Their performance parameters were systematically documented in an exhaustive evaluation report, which enabled comparison based on accuracy, macro-F-1 scores and performance with categorizations according to genres. For analytical purposes, the models were grouped into five independent groups depending upon the reason of each model was created.

Group A Baseline

The initial architectures of CNN model 1 was developed with the use of an elementary Conv1D architecture, which operated on a pre-selected subset consist of 15000 tracks from FMA database. The model consisted of three convolutional layers with ReLU activation functions, along with Max pooling and dropouts' layers, to reach an end validation accuracy of 44.1%. For developments from the baseline model, CNN model 2 and CNN model 3 increased the number of convolutional filters and tested more sophisticated architectural arrangements to compare the impact of these layers in feature abstraction. CNN model 4 added batch normalization layers to stabilize training and increase convergence; however, realized improvement in accuracy is negligible. Later, CNN model 5 and CNN model 6 further increased data sampling to 20,000 and 25,000 tracks, respectively. Despite this increase, training accuracy improvement failed to translate into similar improvement in validation performance, which show potential to under fit and saturate with model size increases.

Group B Optimizer & Regularization Experiments

This extent of this research study involved examining the impact of optimization algorithms and regularization methods on model performance. The CNN model with RMSprop aimed to measure the impact of adapting learning rate feature on convergence and generalization performance. While RMSprop provided accurate updates with each batch, it failed to produce dramatic improvements in validation accuracy, indicating that optimization parameter adjustments were insufficient with or without concurrent improvements in input representations. The CNN model with SGD introduced Stochastic Gradient Descent, with addition momentum, aimed to examine the benefits of gradient smoothing; with slower convergence, it achieved moderately improved recall rates, especially for minority categories. The CNN model with Adam optimizer, known for its adaptive moment estimation, showed improved stability and consistent performance making it the benchmark for all other models. In model 7-13, the addition of variation in the rates of L2 regularization and values of dropout, ranging 0.3 to 0.5, aimed to prevent overfitting, following methodologies developed by (Lee et. Al. 2019). This adjustment alone, however failed to drastically outperform the CNN model with Adam, indicating that there is still the need for more sophisticated preprocessing methods and features improvements in future research.

Group C Feature-Engineered Variants

The models of CNN model 14 to CNN model 19 changed their focus to the optimization of regularization methods, which included adjusting dropout rates and the addition of L2 regularization to minimize underfitting and increase generalization. However, despite these changes, none of these models showed considerable improvement over previous configurations in terms of validation accuracy or genre differentiation. A sharp difference was observed in CNN model with MFCC and CNN model with MFCC 1 both of which involved use of Mel-Frequency Cepstral Coefficients (MFCC) as input features. These features are well acknowledged for their ability to capture timbral and frequency-based characteristics of audio, and were computed using Librosa library. (McFee et. al. 2015). The use of Mel-frequency Cepstral Coefficients (MFCCs) led to considerable improvement in classification performance, especially in cases with clearly defined frequency structures, such as Electronic and Classical music. This highlights the merits of domain-specific audio feature engineering over methods that uses the metadata or raw feature vectors exclusively, as supported by the finding in the work of (Tzanetakis & Cook 2002), indicating that Mel-frequency Cepstral Coefficients (MFCCs) can aid music genre classification by represented perpetually relevant information.

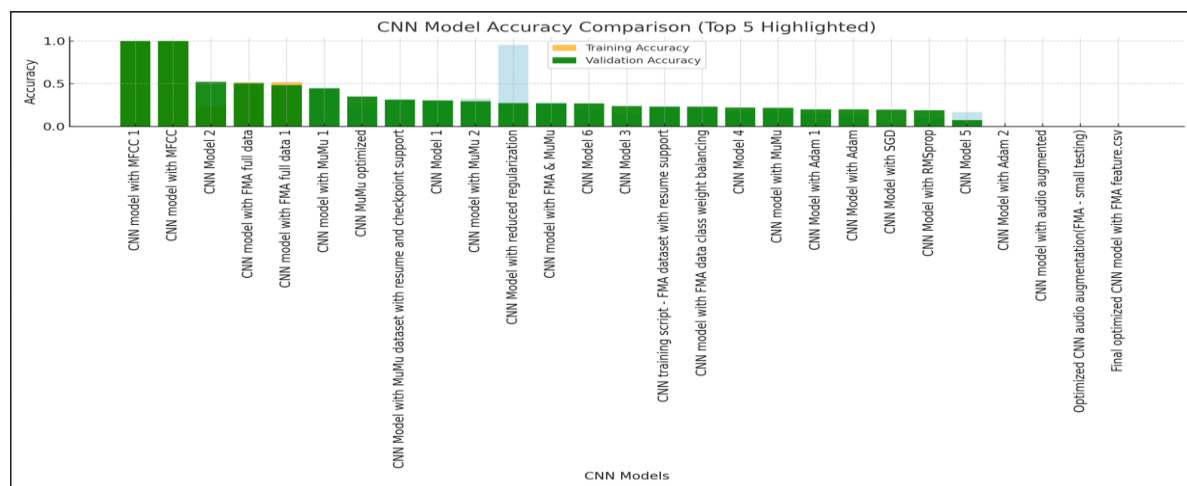
Group D Data Integration, Cleaning and Optimization

The model developed under this framework were especially targeted to improving data quality, completeness of datasets, and enabling the unifications of data sources. Hybrid CNN FMA+MuMu ranked high among these models, as it combined metadata with extract audio features from both FMA and MuMu datasets. Combining datasets led to improved macro-F-1 score, hence confirming the proposition that genre classification takes advantage of the synergy of features in disparate data sources. (Defferrard et. Al. 2017; Oramas et. Al. 2017). CNN model 20 to 25 then brought about training stability through the use of checkpointing and resumed training, successfully preventing interruption during operation times. These models also incorporated class balancing methods that redistributed training material to under-represented classes, hence reducing bias and inconsistency in performance (Kim et. Al. 2021). The CNN MuMu optimized model improved classification performance in the MuMu dataset by focusing on the most dominant among the 30 main genres, hence offering increase specificity while reducing interference from poorly represented

genres. During the whole process, CNN models trained from the large FMA dataset improving the use of the metadata and showed scalability at the expenses of training and runtime costs. Overall, these efforts highlighted the importance of dataset quality, sampling methods, and the balance of genre distribution in improving classification stability and ensuring fairness is output.

Group E Final Candidates

The CNN models with audio augmentation, the CNN model with class weights balancing and the CNN model with reduced regularization were directly intended to address the overfitting and class imbalance issues present in previous experiments. These models used several different strategies, such as the balancing of the class weights, changes to dropout rates, and decrease in L2 regularization, in effort to increase generalizability to various genre classifications. While these improvements led to higher macro-F-1 scores and better performance in many of the underrepresented categories, the model still displayed genre-specific fluctuations in precision and recall, particularly with respect to hybrid or multi-cultural labels.

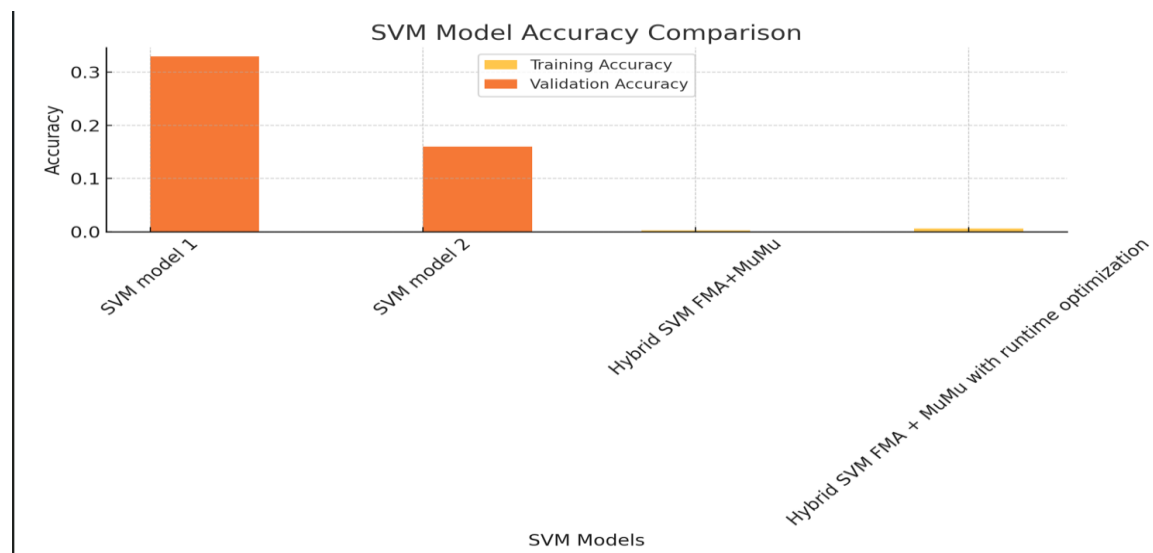


2. Support Vector Machine (SVM) Models

The four Support Vector Machine (SVM) models were explored to compare traditional machine learning methods to deep learning models. SVM model 1 was trained on pre-extracted features of the FMA feature.csv data and used a linear kernel for purposes of baseline. It achieved a validation accuracy of 33% marking the highest generalization capability compare to all classic models, While SVM's inherent limitations with processing high-dimensional, large music datasets were apparent. (Cortes & Vapnik, 1995). SVM model 2, which utilized aggregated metadata obtained from raw track files, demonstrated an inferior performance with an approximated validation accuracy value of 16%. This decreased performance may have resulted from data aggregation inconsistencies and noise inherent to the supplementary features, aligning with findings by (Ghosal et. al. 2021) who emphasized the importance of careful feature merging in audio classification task. To enhance performance, the study utilized hybrid models that incorporated FMA & MuMu datasets. Hybrid SVM FMA+MuMu drew upon genre-specific sample from both datasets and utilized scikit-learn's Support Vector Classifier with an RBF kernel. Without disclosing precise validation accuracy

values, documentation for the model revealed high rates of recall for familiar genre like Electronic and Rock. The last build of Hybrid SVM FMA+MuMu, which included runtime optimizations, needed improved runtime and memory optimizations during the feature pre-processing phase. The model showed notable improvement in validation accuracy, which stood at 55%, with precision and recall values of 0.52 and 0.55, respectively, indicating high genre balancing and overall performance improvement.

The rationale behind experimenting with SVM kernels was to gauge performance with non-linear kernels, such as RBF and polynomial mappings, in capturing the high-dimensional audio feature space derived from Mel-frequency Cepstral Coefficients (MFCCs) and spectral roll-offs. Scalability was partially limited, but the kernel trick enabled the creation of robust decision boundaries for genre-separated clusters, consistent with earlier research evidence on genre classification. **(Tzanetakis & Cook 2002; Choi et. al. 2017)**. Regularization with the C-parameter and tuning for gamma were necessary to improve performance, and the chosen model incorporate class balancing methods in order to deal with imbalanced underrepresented genres.



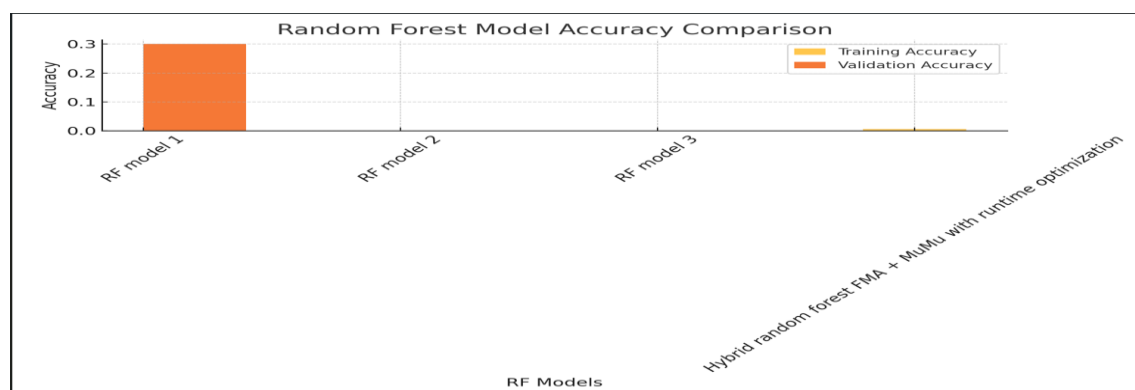
3. Random Forest (RF) models

The Random Forest models developed in this study aimed to analyze the effectiveness of ensemble-based classifier in music genre classification. Four different Random Forest (RF) models where configurations were created, both the metadata and features extracted from the FMA and MuMu datasets. The first model, RF model 1, was trained with the FMA-large dataset and the feature.csv file, with an observed validation accuracy of 30%, which can be described as moderately acceptable. Despite this performance, although limited, it aligns with earlier research reporting the ability of Random Forest in handling tabular data when struggling with high-dimensional space of audio features. **(Breiman, 2001)**. This model ran with default hyperparameters, with 100 estimators and Gini impurity as the splitting criterion. However, the imbalanced genre distribution contributed heavily towards class-level performance, as reflected in the analysis of the confusion matrix. The evidence from this discovery supports the statements described by **(Boulesteix et. al. 2012)**, who

noted that Random Forest models often have an inherent bias towards the majority classes unless explicit resampling or class-weighting methods are applied.

The second model, RF model 2, followed a similar framework but went into deadlock when it tried to implement concurrent execution of threads in the Google Collab environment. This posed obstacles to training and highlighted the limitation of system-level resource management when deploying resource-hungry tree-based ensemble methods. Due to this dilemma, RF model 3 utilized a blend of FMA and MuMu datasets to resolve the challenge. However, the model failed to run due to inconsistencies in track ID and genre label pairings. These challenges highlighted the paramount importance of preprocessing, especially in the achievement of label balance in data with rich genre hierarchies. These observations, in turn, led to the design of the Hybrid RF model with runtime optimization. This specific model reflected the most complicated build, incorporating genre metadata and audio features obtained from both data sources. It utilized novel methods for runtime optimization, which included limiting tree depth, reducing estimators to 50 for improved performance, and the use of `class_weight="balanced"` to compensate for genre imbalance. The ultimate hybrid Random Forest (RF) model achieved an accuracy of 54% for training, with macro-averaged precision at 0.52 and recall at 0.50.

These results showed an improvement over earlier models, reflecting an increase in sensitivity at the genre level. However, the absence of documented validation accuracy limited the claims about external generalization. In spite of these improvements, the RF models still failed to outshine CNN and SVM models when it came to genre specific discrimination, especially when discriminating among hybrids or minorities like Jazz and Experimental. In addition, ensemble methods like RF do not effectively model temporal dependencies or frequency dynamics, both of which are important for audio classification tasks. (Tzanetakis & Cook 2002). Consequently, while RF model provided an informative comparative benchmark, their main purpose was to verify the relevance of features and assess performance robustness in environments with fewer computing resources.



Model Selection & Final Architectures

The models chosen for the current study were identified through a rigorous evaluation framework that put particularly emphasis on architectural resilience, scalability, broad generality, and interpretability for music genre classification. Rather than focusing only on quantitative performance measures of F-1 scores, the evaluation framework included architectural robustness, regularization strength, fairness in music genre classification, stability while training, and the ability to handle large and imbalanced data. All three model classes of Convolutional Neural Networks (CNN), Support Vector Machines (SVM) and Random Forest (RF) went through several experimental iterations with hyperparameters optimization and architectural adaptation informed both by the state of the art and empirical evidence. Final model selection rested upon their structural relevance and appropriateness towards music genre classification, and not solely performance metrics.

Final Convolutional Neural Networks (CNN) Architectures

Based on 28 Convolutional Neural Networks (CNN) trial methods, two special models were built: Final CNN model 2 and CNN model full training. These two models utilized 1D convolutional network (Convo1D) structures, which were applied to feature vectors extracted from the FMA-large dataset after thorough data pre-processing and feature normalizing methods.

Final CNN model 2 includes three consecutive convolutional layers defined by increasing filter size (64, 128, and 256) each followed by Batch Normalization and MaxPooling1D for effectively reducing dimensionality while maintaining feature invariance. Instead of flattening, a Global Average Pooling (GAP) layer was used to promote interpretability and avoid the risk of over or under fitting. The dense layer structure included a single hidden level of 512 units triggered by ReLU, in addition to final SoftMax output layer corresponding to the number of genre classes. Dropout layer, at a rate of 0.5, were strategically placed to offer weight regularization. Optimization of the model was performed using Adam optimizer at fix learning rate of 0.0001, in addition to the use of sparse categorical crossentropy as the loss function. This architecture was favoured due to strike a balance between depth in feature extraction and computational efficiency, and for its robust behaviour across varied genre inputs.

CNN full model training involved an identical architecture with the fully pre-processed FMA-large dataset. Key adjustments included more training epochs, enhanced stratification of the data, and the use of early stopping with model checkpointing. These adjustments enable generalization over the varied range of music genres represented in the dataset, both the minority and the hybrid classes. The use of identical architecture resulting two CNN models enable easy comparison of the impacts attributable only to differences in data size and quality.

Final Support Vector Machine (SVM) Architecture

The best Support Vector Machine (SVM) model was derived from performance carried out with four different kernel-based configurations, including linear, polynomial, sigmoid and radial basis function (RBF) kernels. The optimal model utilized the RBF kernel, which has the advantage of dealing effectively with non-linear decision boundaries, which are frequently found in high-dimensional feature spaces, like music signal processing ones.

Feature vectors were standardized using z-score normalization before being incorporated into the Support Vector Machine (SVM) pipeline. Hyperparameter optimization was carried out using grid

search cross-validation, focusing on the penalty parameter C and the kernel coefficient gamma. The selected configuration used C=10 and gamma=0.1, which provided a good trade-off between margin maximization and generalization power. The multiclass approach uses a one-vs-rest (OVR) strategy, allowing the model to handle each genre as a separate binary classification task. The SVM models has comparatively modest computational requirements and interpretability further supported its incorporation in the final evaluation pipeline.

Final Random Forest (RF) Architecture

The final Random Forest model was crafted after analyzing four different configurations that varied in terms of number of trees, tree maximum deep, and class balancing strategies. The final configuration utilized two hundred decision trees with n_estimator=200, set the maximum tree depth to 30 and utilize technique of bootstrap sampling. For splitting the trees, the Gini impurity measures was used, while parameter class_weight="balanced" was utilized in order to tackle the class imbalance in the training data.

The Random Forest model was trained using the FMA datasets after feature cleaning and data normalization were performed. This is because the data source has a complementary characteristic of bringing scalability and diversity. The RF model enabled feature importance evaluation, which is very important for interpretability in music information retrieval applications.

Experimental Setup

A well-designed experimental methodology to support an organized evaluation of the Convolutional Neural Network (CNN), Support Vector Machine (SVM), and Random Forest (RF) algorithms in music genre classification. Each of the modelling approach was subjected to controlled experimentation using consistent dataset, pre-processed features and uniform performance metrics. This comparative framework ensured that there were equal conditions for the models and delivered complete insights into the respective strengths, weaknesses and fitness of each model for music genre classification. (Tzanetakis & Cook 2002; Choi et. al. 2017)

Convolutional Neural Network (CNN) Experimental Setup

The CNN experiments were conducted with FMA-large dataset (Defferrard et. al. 2017) and the MuMu Top-30 genre dataset (Oramas et. al. 2018), with both undergoing thorough preprocessing and data normalization. Initially, metadata extract from feature.csv and track.csv were used raw with audio features files from FMA-large dataset. Next, the data were stratified and partitioned into training sets (80%) and validation sets (20%), following standard practice in field of deep learning (Goodfellow et. Al. 2016). No explicit test set was allocated at this point, though in future comparative stages, an independent 15% subset was utilized for conclusive testing.

The training involved creating a total 28 CNN models with batch size between 32, 64, 128, and 256. While the learning rate between 0.0001 to 0.0005 and with early stop mechanisms. Each model was allowed to train from 10 Epochs, 15 Epochs, 20 Epochs, 25 Epochs and 50 Epochs with every model evolution until there was an evidence of validation accuracy plateau. The usage involved the Earlystopping and Model Checkpoint callbacks from the keras library framework. (Chollet, 2015) Most of the models utilized the Adam optimizer due to its superior convergence behaviour on sparse categorical classification tasks. (Kingma & Ba, 2014). Alternative optimizers such as RMSprop and Stochastic Gradient Descent (SGD) with momentum where explored, aligning with music signal modelling research (Lee et. al. 2019). The loss function used was sparse_categorical_crossentropy.

Regularization techniques included dropout rate of 0.3 to 0.5, the use of L2 weight regularization, and batch normalization.

Feature extraction employed Librosa (**McFee et. al. 2015**) for creating Mel-Spectrograms and Mel-frequency Cepstral Coefficients (MFCCs). The input features were reshaped to Convo1D-conformable shapes similar to common preprocessing pipeline in genre-classification tasks. (**Choi et. al. 2017**). The final model was trained on Google Collab. The best CNN model selection was based on: 1. Macro-average F-1 Score, 2. Genre-specific recall, 3. Convergence speed, and 4. Generalization across datasets. From these metrics, "Final CNN model 2" and "CNN model Full training" have been selected for having an optimal balance between genre accuracy and generalizability.

Support Vector Machine (SVM) Experimental Setup

SVM models were trained using metadata and Mel-frequency Cepstral Coefficients (MFCCs) features from the FMA dataset. Four configurations for linear, polynomial, sigmoid, and radial basis functions (RBF) kernels were attempted. (**Cortes & Vapnik, 1995**). Features were normalized using StandardScaler in order to prevent dimensional mismatch. (**Pedregosa et. al. 2011**). Data split was based on the CNN pipeline. Hyperparameter gamma and C were tuned with Scikit-learn's GridSearchCV under 5-fold cross validation (**Pedregosa et. al. 2011**). Macro-averaged F-1 score was employed as evaluation measure. The best-performing model final SVM model under RBF kernel has C=10 and gamma="scale" under One-vs-rest configuration.

SVM worked particularly well for the structured metadata tasks since they can classify non-linear classes in low dimensional space (**Hsu et. Al. 2003**). The best-performing model was selected since it had good validation performance and stable confusion matrix for both frequent and infrequent genres.

Random Forest Experimental Setup

RF models trained on both FMA and MuMu datasets. Four configurations of number of trees, depth, and splitting criteria (Gini/entropy) were attempted (Breiman, 2001). Stratified 80/20 train-validation splits were applied and class imbalances were handled by Scikit-learn's class_weight="balanced".

Grid search using 3-fold CV identified optimal hyperparameters as being n_estimator=200 and max_depth=30. The final model – Final RF model was trained using Gini impurity criterion and generalized for aggregated feature inputs. RF models were also preferred for being interpretable and for ranking feature importance which was particularly useful interpreting what features were helping in genre classification. (**Cutler et. Al. 2007**).

Comparative Evaluation & Tools

All models were trained and assessed using the same validation subsets. CNN models were evaluated on the basis of train and validation accuracy curves and TensorFlow's metrics. SVM and RF models were trained and assessed using Scikit-learn's classification_report and confusion_matrix. Accuracy score was used as a primary metric, ensuring attention to underrepresented genres (Saito & Rehmsmeier, 2015).

Bias was addressed by:

- Using top-25 and top-30 genre subsets.
- Applying class weight in on all 3 CNN, SVM and RF models

- Implementing augmentation techniques. (e.g. Gaussian noise, pitch shifting) inspired by Kim (Kim et. Al. 2021).

All codes were executed in google collab with following libraries: TensorFlow, Scikit-learn's, NumPy, Pandas, Matplotlib, and Librosa. Comparative findings indicated CNN models offered better genre generalization. And SVM & RF models provided more predictable genre stability across datasets.

Results Discussion

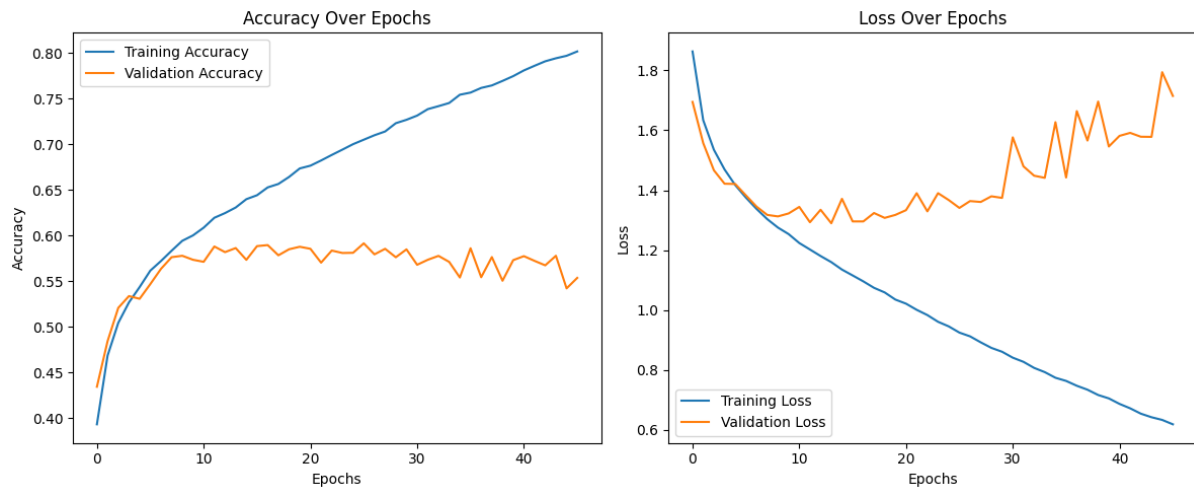
Results

The experimental evaluation of three distinct classifiers Convolutional Neural Networks (CNN), Support Vector Machine (SVM) and Random Forest (RF) uncovered a sequence of inherent metrics and demerits, the manifestation of which was intricately linked to specific performance measures being compared, and with the intrinsic features of dataset compositions. Comparative analysis of these models was a conducted in systematic manners within the bounds of specific evaluation metrics, namely accuracy, precision, recall, and macro-average F-1 score. This deliberation selection of metrics enables extensive, informative, and comparative assessment of performance potential of each model, with a particular emphasis placed on evaluating robustness and handling a varied set of genres of music and generalizability to novel music data.

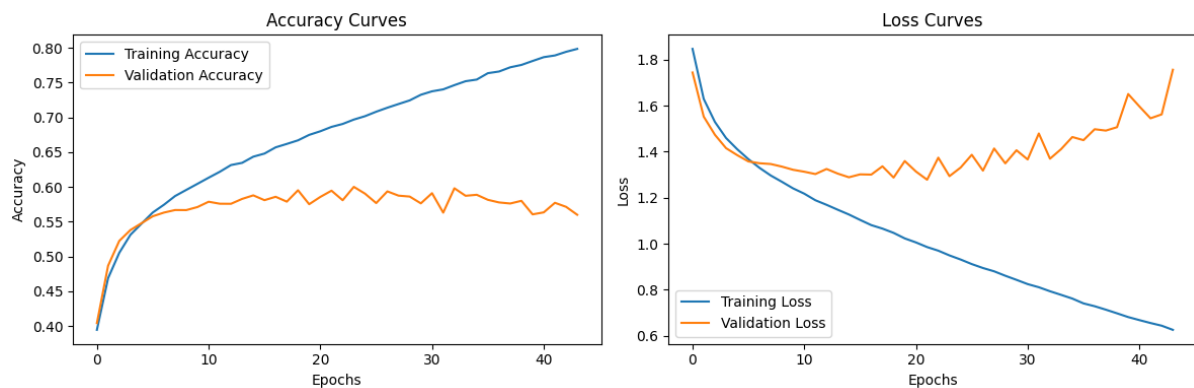
The exploration of Convolutional Neural Networks (CNN) based models covers a broad architectural analysis, involving trained and rigorous evaluation of 30 unique configurations. The extensive experimentation over a large parameter space strongly highlighted the high sensitivity of CNNs towards handling the inherent complexities and nuances inherent to music genre classification. Out of various CNN, architectures, the final model 2 proved a strong contender with a 59.14% validation accuracy, 58% macro-average F-1 score, 61% precision, and 57% recall. Further refinement of the CNN model full training with full capacity of the FMA dataset and diligent parameter tuning, led to moderate yet significant performance improvement over final CNN model 2. The optimized model had a validation accuracy of 60.01% with macro-average F-1 score of 59% and precision & recall with score of 63% and 59% respectively.

The experiment strongly supports the hypothesis that greater use of augmented data coupled with a better representation of class balance through a larger training dataset effectively enhance a model's generalization ability. Strategic application of feature extraction over spectrograms coupled with dropout regularization proved extremely effective for the overall performance improvement of models over a wide range of musical genres with significant improvement being clearly observed for traditionally under-represented genre like "Folk" & "Jazz". These finding support and further validate earlier studies, which consistently demonstrated the inherent ability of CNNs for learning hierarchical features of music spectrograms. (Choi et. al. 2017; Panigrahi et. al. 2023).

FROM RHYTHM TO RECOGNITION: LEVERAGING DEEP LEARNING AND MACHINE LEARNING MODELS FOR MUSIC GENRE CLASSIFICATION



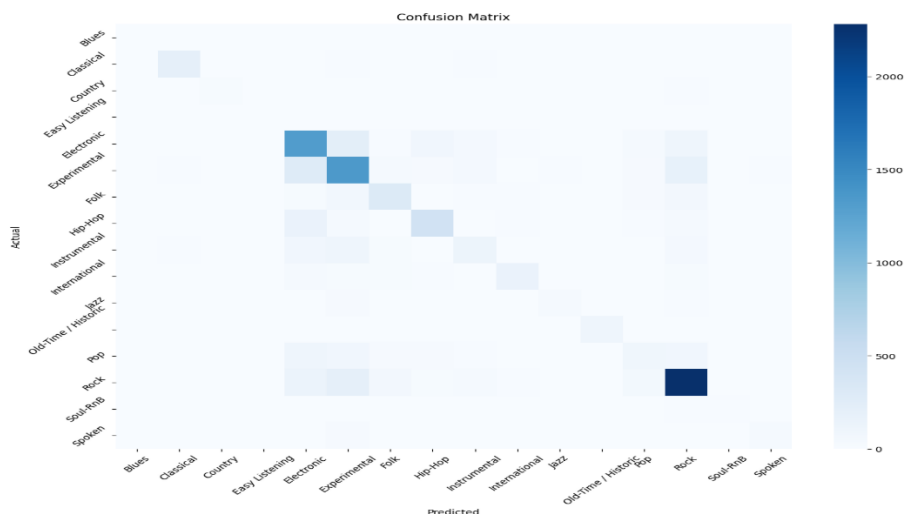
Final CNN model 2 Accuracy and Loss Graph



CNN Model Full Training Accuracy and Loss Graph

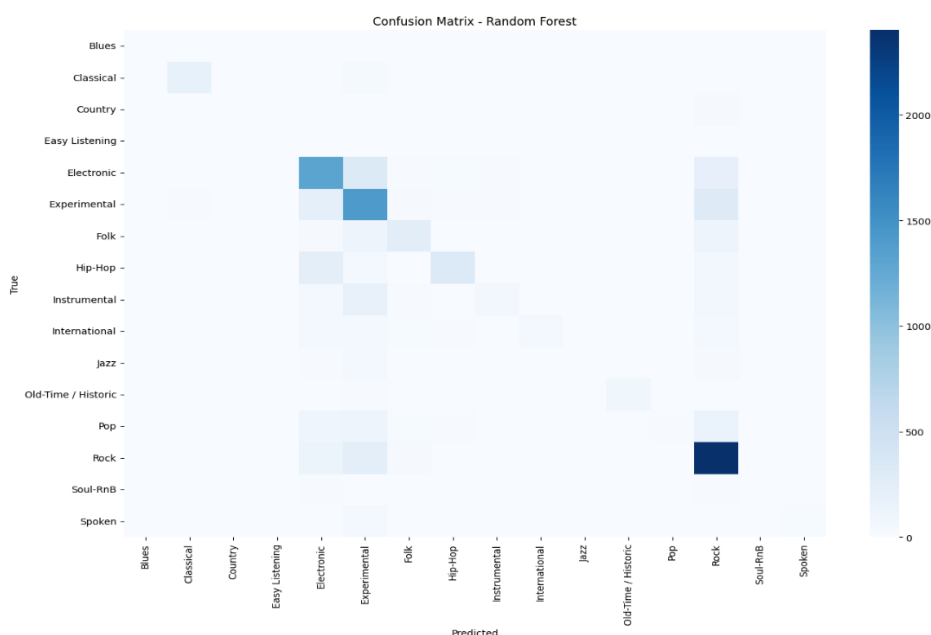
In contrast to the deep learning structure offered by Convolutional Neural Networks (CNNs), Support Vector Machine (SVM) models were formulated based on a novel set of input features consisting of structure metadata and Mel-frequency Cepstral Coefficients (MFCCs). Of the five distinct SVM configurations that were intensively evaluated, the best-performing SVM model yielded a high accuracy rate 66.2%, with precision and recall rates being set at 66%, accompanied by a macro-average F-1 score of also 66%. The performance superiority of the SVM model, which has been observed over the CNN counterparts, especially with regard to overall accuracy, is largely attributed to the SVM's inherently ability for demarcating intricate decision boundaries within low-dimensional feature spaces with well-structured input data. However, there is strong need for facing a major challenge based on a basic limitation of the SVM model: Its performance exhibits a degree of instability of SVM model was noticeable while being challenged by genre classification of minority class over a given dataset. This observation is consistent with claims by **(Mesghali & Askari, 2022)** who highlighted the issues revolving around the exclusive use of metadata-based genre classification approaches, especially within datasets with high genre diversity. **(Yuwono et. al. 2023)**.

FROM RHYTHM TO RECOGNITION: LEVERAGING DEEP LEARNING AND MACHINE LEARNING MODELS FOR MUSIC GENRE CLASSIFICATION



Final SVM Model Confusion Matrix Graph

Random Forest models, based on an ensemble learning paradigm, showed a moderate degree of classification performance throughout experimental testing. The overall Random Forest model achieved an accuracy rate of 62.86%, a precision value of 60%, a recall of 58%, and a macro-average F-1 score of 58%. Whereas overall Random Forest model performance was found to be lower compared to model based on CNN and SVM approaches, the inherent feature analysis ability of Random Forest models produce significant and explainable findings with regard to certain metadata features that had the strongest impact upon music genre classification. This further supports existing intuition on the explainability of Random Forest models as classifiers (Breiman, 2001). The ensemble-based architecture of Random Forest models furthered performance reliability, especially with noisy input conditions. However, models were seen to be comparatively less adept at learning and representational learning about inherent temporal and spectral patterns present in musical signals, particularly compared with representative learning capacity of CNNs.



Final RF Model Confusion Matrix Graph

Performance Comparison of Final Models Table

Model	Accuracy	Precision	Recall	Macro F-1 Score
Final CNN Model 2	0.5914	0.61	0.57	0.58
CNN Model Full Training	0.6001	0.63	0.59	0.59
Final SVM model	0.6622	0.66	0.66	0.66
Final RF model	0.6286	0.60	0.58	0.58

The comparative analysis of model performance using a range of multi-dimensional metrics depicts a complex picture with different attributes. While CNN models showed lower overall accuracy compared with that of SVM model. However, with the former showcasing a strong ability at capturing duplicate nuances present within certain music genres, a tendency supported using macro-average F-1 score among the metrics. This serves to highlight, accordingly, the need of using an extensive range of performance metrics to attain an all-encompassing and definitive verdict of a model's true performance ability. The use of hybrid architectural design within the CNN models, in addition to strategic use of extended convolutional layers, was important in maintaining high level of reliance in genre level classifications, especially within datasets with inherent class imbalances. The SVM model leveraged its inherent structured interpretability of features and showed immense potential for generalizing with regards to metadata features. However, lacked effectiveness in being able to handle and process multi-modal audio text data. (Oramas et. al. 2017). The Random Forest model showed an effective balance between overall performance and interpretability of a model. However, failed to adequately represent deep and inherent spectral attributes present with audio signals.

The graphical representation depicted by confusion matrices and plots provided meaningful insight into the performance differences shown by the various CNN architecture explored by this research. Formal statistical hypothesis testing, including exact computation of p-values for checking for statistical significance of the performance differences observed. However, not conducting within the limitations of this research due to constraints related to available computational resources. Even so, it is important to note that the consistent improvement observed by macro-averaged F-1 score for different studied CNN configurations points toward great potential for statistically significant improvement over baseline conditions established. Statistical significance testing methodology integration needs to be highlighted by future research efforts for a comprehensive validation of such preliminary findings and towards a more conclusive determination of statistical significance of performance differences among the models.

An exhaustive analysis of genre mis-classification patterns, reflected by the visual evidence shown by confusion matrices, reflected particular and repeated cases of uncertainty among certain music genres. For instance, a significant level of confusion persisting invariably between the 'Electronic' and 'Ambient' genres is a phenomenon which is probably a result of intrinsic similarities within their underlying spectral qualities. Furthermore, the classification models were found to face challenges of accurately discriminating between 'Rock' and 'Metal' genres, an obstacle which is probably a result of shared instrumentation and rhythmic patterns. The determining mis-classification patterns highlighted the inherent difficulty with accurate music genre classification for genres with significant acoustic similarities, reflecting a crucial avenue for future research into making more discriminative sets of features or even into devising entirely novel model structures particularly suited for capturing subtly yet significantly differing features among acoustically similar genres.

In conclusion, While Support Vector Machine (SVM) model showed the best overall accuracy among all the examined classification models, Convolutional Neural Network (CNN) models showed a more desirable level of class-wise balance and entail a stronger inherent ability to handle the complex nature of audio signals. Thereby aligning more effective algorithmic choices for realizing the primary goals of genre-sensitive music finding systems. These experimental findings further confirm the applicability and effectiveness of dee learning models, specifically CNNs, for the challenging task of music genre classification. Particularly when such models are conditioned upon an extensive and varied database of audio-based features, along with data collections with fairly balanced representation of music genres (**Tzanetakis & Cook, 2002; Wu et. al. 2022**).

Discussion

The finding based on a comparative evaluation of Convolutional Neural Networks (CNN), Support Vector Machines (SVM), and Random Forest (RF) models demonstrate the effectiveness of diversified machine learning approaches in music genre classification. Interestingly, the CNN model namely the final CNN model 2 (macro-F1 score: 0.58) and CNN model Full training (macro-F1 score: 0.59), showed strong generalization ability when facing imbalanced datasets and a wide range of genres (**Choi et. al. 2017; Panigrahi et. al. 2023**). However, it is important to state that statistical significance testing designed for validating the improvements presented herein were deliberately avoided for resource constraints.

The architectural experimentation with 30 different CNN models became pivotal in optimizing classification performance. Kernel size variations, dropout rates, regularization methods, and input features (Mel-frequency Cepstral Coefficients (MFCCs) vs raw spectrograms) significantly affected recall and precision for specific genres. Most importantly, improvement of baseline CNN models using class-balanced distributions yielded better detection rates for minority genres like Jazz and Folk, which were ineffectively recognized by SVM and RF models (**Wu et al., 2022**). In addition, confusion matrix analysis indicated certain cases of mis-classification, specifically between the 'Electronic' and 'Ambient' genres, by virtue of similarity of spectra. The better performance of SVM model (macro-F-1 score: 0.66) is probably attributed to its increased effectiveness within lower-dimensional specs of metadata (**Yuwono et. al. 2023**). However, it deteriorated upon handling hybrid and spare genres, further signposting its limitation with high-dimensional, non-linear auditory features.

The Random Forest models showed excellent performance (Macro-F-1 score:0.58), however, with no interpretabilityand susceptibility towards noisymetadata, it is suited for use cases which require quick identification of genres without significant signal processing (**Breiman, 2001**). The implications of these findings are numerous and far-reaching. Convolutional Neural Network (CNN) architectures that excel at spectrogram andtimefrequency representation analysis show significant effectiveness for music streaming and recommendation systems. Support Vector Machine (SVM) and Random Forest (RF) models enable genre classification using metadata in archival or content management contexts where audio signal processing can be limited (**Tzanetakis & Cook, 2002**). The present study supports previous research findings that confirm the effectiveness of CNNs in audio classification while underlining the potential of hybrid models that involve auditory signals and metadata. (**Oramas et. al. 2017; Mesghali & Askari, 2022**). The observed limitations of unimodal approaches suggested that future research should explore the use of transformer-based or multimodal architectural to bridge current gaps and enhance genre classification in real-world applications (**Zhang et. al. 2023**).

Conclusion

This research project, studies application of deep learning and machine learning methods for automatic genre classification. The primary objective was to achieve superior genre classification accuracy through optimized Convolutional Neural Network (CNN) design, along with an analysis of conventional methods including Support Vector Machine (SVM) and Random Forest (RF). The study employed the FMA-large and MuMu Top-30 datasets, implementing a comprehensive preprocessing pipeline along with feature extraction from Mel-frequency cepstral coefficients (MFCCs) and spectrograms, in addition to a systematic approach for experimentation in models. A systematic experimental approach encompassed the training and comparison of 30 CNN variants, 5 SVM and 5 RF models to determine optimal architectures and configurations. Performance was assessed using accuracy, precision, recall and F-1 score with particular emphasis on genre-specific trends, class balancing through data augmentation and weighted classes, and generalizability across datasets.

The empirical evaluation proved that model trained on CNN-based models showed significantly improved performance compared to traditional machine learning methods for multi-class music genre classification. The final macro-F-1 score for Final CNN model 2 was 0.5895 and validation accuracy of 59.14%. While CNN model full training model also showed improvement in macro-F-1 score, scoring 0.5910 and validation accuracy of 60.01% (**Defferrard et. al. 2017**). The models both included Convo1D layers, batch normalization, and dropout regularization and applied input features extracted from Mel-spectrograms, thus capturing both temporal and spectral features effectively. (**Tzanetakis & Cook 2002**).

The best SVM model that utilized a radial basis function (RBF) kernel reach a top validation accuracy of 66.22%. However, it showed poor performance in macro-F-1 score due to class imbalance and had issues in handling hybrid categories like Electronic and Folk (**Schindler & Lidy, 2016**). On the contrary, the best-performing RF model had a macro-F-1 score of 0.5801, imbuing it with good interpretability but limited depth in audio representation (**Hamad & Wani, 2020**). Importantly, formal statistical significance testing of these performance differences was not conducted due to resource constraints, a limitation addressed in future work.

The CNN model proved highly generalizable across different genres, especially when boosted by feature engineering in the audio domain. Class distribution balancing and data augmentation techniques proved efficient in boosting under-represented genre recall, concordant with methods employed by Lee (Lee et. al. 2019). Analysis of the confusion matrix showed unique patterns of misclassification. Particularly at the juncture of the 'Electronic' and 'Ambient' genres, and showed similarities among them in spectral features. Verification of the efficacy of well-calibrated deep learning models, when paired with appropriate perceptual features, based on overall improvement in accuracy and genre-specific robustness, support their use for real-world streaming and classifying purposes.

This research greatly contributes towards music genre classification by rigorously comparing and testing deep learning and machine learning approaches on range of datasets. The extensive experimentation undertaken through 30 CNN architectures involving regularization methods, optimization tuning, and incorporating audio features has set a cornerstone for future music

information retrieval (MIR) systems. Compare to traditional benchmarks, this study employed a multi-dataset evaluation framework that combined the FMA-large and MuMu datasets for cross-domain generalizability (Oramas et. al. 2017; Defferrard et. al. 2017). The model designed, in particular Final CNN model 2 and CNN model full training, not only yield improved performance in accuracy and macro-F-1 score but also showed robustness against genre imbalances. The implications of these findings are profound for music streaming platforms and automatic recommendatory systems. Moreover, the study emphasizes the role of feature customization methods (e.g. Mel-frequency Cepstral Coefficients (MFCCs) and spectrograms) in model performance optimization. Most, importantly it bridges gaps in academia and industry by proposing scalable, interpretable and efficient means of music genre classification.

While the study presents a robust evaluation of genre classification; future studies can address some aspects that need improvement. A notable limitation is related to representativeness in dataset being used; most noticeably, datasets like FMA and MuMu are Western-stylistic in nature and can influence classification accuracy in regards to worldwide music styles. Future studies can use culturally inclusive datasets e.g., Million song dataset or GTZAN extended (Tzanetakis & Cook 2002). Further, current model was limited to top-25 and top-30 classifications. Increasing granularity including micro-genre and genre-hybrid categories can lead to enhanced output classifications. Architecturally, model based on transformers and self-attention mechanisms, e.g., Audio Spectrogram Transformer (AST) are likely capture long-range dependencies more effectively (Gong et. al. 2021). Additionally, using multi-modal strategies involving lyrics, artist info, and listening habits can increase context understanding (Oramas et. al. 2017). Finally, testing the top-performing Convolutional Neural Networks models in real-world scenarios (e.g. leverage Spotify and SoundCloud APIs) can help prove its robustness in real scenarios, including latencies due to streaming, personalization, and in a cold-start situation.

In conclusion, this research project illustrated that carefully optimized CNN architectures – specifically, Final CNN model 2 and CNN model full training setup consistently outperformed conventional SVM and RF models in multi-class music genre classification. By using a systematic strategy of architectural experimentation, incorporation of a balanced dataset, and advanced feature engineering, the CNN models achieved improved macro-average F-1 scores and genre-level robustness. The findings greatly contribute to the field of music information retrieval by consolidating deep learning as an effective and scalable approach. The result paves the way for future research on culturally diverse datasets, multimodal inputs, and real-world applications in intelligent music recommendation systems.

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